

Lukáš Randuška

Data Science · Space Weather Research

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PROFILE

Undergraduate student in Data Analysis and Artificial Intelligence with a research focus on the intersection of data science and space physics. Currently a research intern at the Institute of Experimental Physics of the Slovak Academy of Sciences, working on space weather data analysis and the prediction of geomagnetic storms and coronal mass ejections (CMEs). Co-author of two peer-reviewed publications in *Earth and Space Science*. Strong foundation in mathematics, scientific computing, and electronics, with the goal of building professional experience in space technology and physics-driven data research.

EDUCATION

Pavol Jozef Šafárik University in Košice

2024 – present

Undergraduate studies, Data Analysis and Artificial Intelligence — Košice, Slovakia

Secondary Vocational School of Electrical Engineering, Košice

2020 – 2024

Department of Information and Network Technologies — Košice, Slovakia

- Completed with the *maturita* (secondary school-leaving examination).
- Final-year project and SOČ (Secondary School Professional Activity) advanced to the national finals in 2023/2024.

RESEARCH EXPERIENCE

Research Intern

Jul 2023 – present

Institute of Experimental Physics, Slovak Academy of Sciences — Košice, Slovakia

- Perform data analysis of space weather and develop prediction models for geomagnetic storms.
- Estimate the propagation speed and onset times of coronal mass ejections (CMEs).
- Study solar wind behaviour and the topology and structure of CMEs.
- Apply Python-based scientific computing (NumPy, pandas, SciPy) to large heliophysics datasets.
- Contributed to research resulting in two co-authored peer-reviewed publications.

PUBLICATIONS

Co-author of the following peer-reviewed articles in *Earth and Space Science* (AGU):

- Mackovjak, Š., Kostárová, S., Chum, J., et al. (2026). "Dataset of Quiet Space Weather Periods for Ionospheric Studies." *Earth and Space Science*, 13, e2025EA004537. DOI: [10.1029/2025EA004537](https://doi.org/10.1029/2025EA004537)
- Kisvárdai, I., et al. (2025). "Analysis of 42 Years of Cosmic Ray Measurements by the Neutron Monitor at Lomnický štít Observatory." *Earth and Space Science*, 12, e2024EA003656. DOI: [10.1029/2024EA003656](https://doi.org/10.1029/2024EA003656)

PROJECTS

MWP — Music Without Pain

2023/2024

Final-year project & SOČ competition entry

- Software project developed as final-year and SOČ work; advanced to the national finals of the SOČ competition (2023/2024).
- Source code: github.com/peto59/MWP

TECHNICAL SKILLS

Programming: Python (proficient — NumPy, pandas, SciPy); Java (intermediate — two semesters of algorithms, data structures, and dynamic programming).

Focus areas: scientific computing, data analysis, space weather / heliophysics data, electronics.

LANGUAGES

Slovak (native) · **English** (B1) · **German** (Deutsches Sprachdiplom — DSD Pro I)

AWARDS & ACTIVITIES

Hack Košice — Hackathon

2021, 2022, 2023, 2026

- 2021 — 2nd place overall. · 2022 — 1st Sponsor Prize (Telekom).